



ENTERPRISE FINANCE INTELLIGENCE

DClaw Finance

AI-powered financial management for enterprise teams — built on the One Convergence design system, powered by DKube private AI infrastructure.

v1.2 · May 2026

11 AI Features

FastAPI · Next.js · PostgreSQL

TODAY'S SESSION

What we'll cover

A walkthrough of the DClaw Finance platform — from product vision to technical architecture, AI capabilities, live financial data, and deployment.

01 → 03 Problem & Solution	04 → 05 Architecture & Stack
06 → 07 AI Feature Set	08 → 09 Live Financial Data
10 → 11 Client Portfolio	12 → 15 Deployment & Roadmap

THE PROBLEM

Enterprise finance teams operate blind.

01

No AI in the workflow

Expense categorisation, receipt parsing, and anomaly detection are done manually — slow and error-prone.

02

Fragmented tools

Invoices in one tool, expenses in another, forecasts in spreadsheets. No single source of truth.

03

Reactive, not predictive

Leaders see last month's P&L two weeks late. Cash flow crises arrive without warning.

THE SOLUTION

One platform. All your finance data.

DClaw Finance is a full-stack enterprise financial management platform with AI built into every workflow. It runs on your infrastructure — private, compliant, and fully controlled.

- Invoice management
- Expense tracking
- AI categorisation
- Cash flow forecast
- Monthly reports
- NL Chat

P0 – MUST HAVE

AI expense auto-categorisation · Monthly trend chart · Invoice reminder drafts · Receipt OCR

P1 – SHOULD HAVE

Cash flow forecast · Anomaly detection · Monthly summary report · Line-item suggestions

P2 – DIFFERENTIATORS

Natural language Q&A · Budget AI guardrails · Client profitability scoring

TECHNICAL ARCHITECTURE

Built to run anywhere.

A clean three-tier architecture with async Python backend, server-rendered React frontend, and PostgreSQL — deployable on Docker Compose, Kubernetes, or bare-metal.

- Backend port 8096
- Frontend port 3007
- PostgreSQL (dclaw_finance)

FRONTEND

Next.js 14 · App Router

Tailwind CSS · shadcn/ui · Recharts
TypeScript · Server-side rendering

BACKEND

FastAPI · Python 3.11

SQLAlchemy 2.0 async · Pydantic v2
Repository pattern · Alembic migrations

DATABASE

PostgreSQL 16

async pg driver · UUID PKs
Alembic versioned migrations

AI PROVIDER

OpenRouter · Anthropic

claude-haiku-4-5 · claude-sonnet-4-6
Pluggable via single client factory

DEPLOY →

Docker Compose

Kubernetes

Helm Chart

python:3.11-slim

node:20-alpine

AI FEATURE SET · V1.2

11 AI capabilities. All in production.

Every AI endpoint accepts `?dry_run=true` for safe UI development without token spend. LLM costs cut 80%+ via caching.

P0 tasks	claude-haiku-4-5
P1–P2 tasks	claude-sonnet-4-6
Provider	OpenRouter / Anthropic

	Expense auto-categorisation 600ms debounce on description + vendor input → suggested category + confidence badge.	P0
	Receipt OCR Drag-and-drop receipt image → claude vision extracts vendor, amount, date, category.	P0
	Invoice reminder drafts One click on a sent/overdue invoice → AI-written email with editable subject + body.	P0
	Cash flow forecast 6-month exponential smoothing ($\alpha=0.3$) → 3-month projection with confidence bands.	P1
	Anomaly detection z-score > 2 flagging per category. Batch LLM explanation. 1h in-memory cache.	P1
	Natural language Q&A Agentic tool-use loop with 4 DB query tools. "What did we spend on AWS last quarter?"	P2

MORE AI CAPABILITIES

Every page is AI-augmented.

Beyond the core features, AI is woven into discovery flows — suggesting line items as you type, scoring clients before you ask, and flagging budget breaches the moment they cross 80%.

LLM CLIENT FACTORY

Single `llm_client.py` abstraction — swaps Anthropic SDK ↔ OpenAI SDK for OpenRouter transparently. No service-level changes required.



Line-item suggestions

After first line item → 3 suggested additions based on client history.



Monthly summary report

Revenue, top cost drivers, 3 actionable recommendations. claude-sonnet.



Budget AI guardrails

Triggers only on breach ≥ 80%. One specific cut action per category.

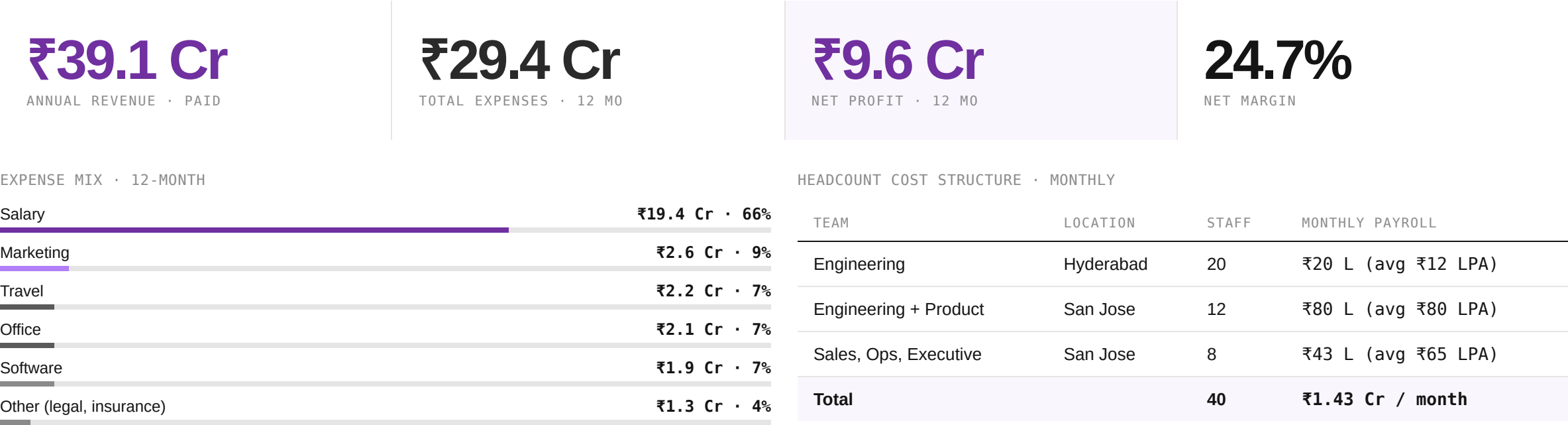


Client profitability score

Composite score (revenue 70% · outstanding 30%). 24h batched insights.

LIVE DEMO DATA • MERIDIAN AI TECHNOLOGIES FY 2025–26

Financial performance at a glance.



INVOICE PIPELINE · FY 2025–26

50 invoices · ₹39+ Cr total value.

42

Paid

4

Sent – awaiting payment

2

Overdue (TIAA ₹2.5Cr · VMware ₹75L)

2

Draft – current month

SAMPLE INVOICES – ENTERPRISE PRICING

INVOICE	CLIENT	SERVICE	AMOUNT	STATUS
INV-2025-001	Apollo Global Management	DKubeX Annual + SLA	₹2.75 Cr	Paid
INV-2025-015	JPMorgan Chase	DKubeX Annual + SLA	₹2.75 Cr	Paid
INV-2025-009	MetLife Group	MLOps Annual + SLA	₹1.75 Cr	Paid
INV-2026-035	TIAA Financial Services	DKubeX Annual	₹2.50 Cr	Overdue
INV-2026-048	JPMorgan Chase	Querilynx Blueprint	₹75 L	Sent

CLIENT PORTFOLIO · 12 ENTERPRISE ACCOUNTS

Financial services, life sciences, tech, legal.

JPMorgan Chase Investment banking ₹7.35 Cr Score 89.8	TIAA Financial Financial services ₹7.05 Cr Score 67.1	Apollo Global Private equity ₹5.75 Cr Score 84.8	Altos Labs Life sciences ₹5.45 Cr Score 81.9
Allianz Life Sciences Insurance · Life sci ₹4.35 Cr	MetLife Group Insurance ₹3.90 Cr	VMware by Broadcom Cloud infrastructure ₹3.65 Cr	Cisco Systems Enterprise technology ₹2.80 Cr

Also:

BuildFirst Construction

StackPath Technologies

Sequoia Legal Partners

Pacific University System

DESIGN SYSTEM

Built on One Convergence Vol. 01.

The UI implements the OC design token system directly — purple ``#7030A0``, paper ``#FFFFFF``, Manrope display face, Inter for body, JetBrains Mono for labels and code. Sharp 2–4px radius, thin 1px rules, no decorative shadows.

Purple 500
`#7030A0` · primary brand

Purple 300
`#B180F8` · accent, hover

Ink
`#141414` · primary text

Paper
`#FFFFFF` · backgrounds

TYPOGRAPHY SCALE

Display `Manrope 800` · headlines

H2 Heading `Inter 600`

Body text — clear, direct, no jargon. `Inter 400`

LABEL / EYEBROW `JetBrains Mono 400`

COMPONENT TOKENS

Primary button

Outline

Feature tag

Active

Status

DEPLOYMENT

Runs on Docker. Scales on Kubernetes.

Three-container Docker Compose for local and staging. Production on Kubernetes with a packaged Helm chart. Non-root containers, health checks, and full environment variable injection from `.env`.

QUICK START

```
docker compose up -d --build
docker compose exec backend \
    alembic upgrade head
```

BACKEND CONTAINER

python:3.11-slim
Non-root appuser
Health check via urllib
Port 8096

FRONTEND CONTAINER

node:20-alpine
Multi-stage build
ARG NEXT_PUBLIC_API_URL
Port 3007

DATABASE

postgres:16-alpine
Named volume persistence
Alembic migrations
Port 5434 (host)

KUBERNETES / HELM

Chart in helm/dclaw-finance/
Secrets via K8s secrets
Namespace-isolated
Rollout status checks

DEMO DATA PROFILE

Meridian AI Technologies

A fictional enterprise private AI company modeled on **DKube** — 40 employees across San Jose CA and Hyderabad India. The demo data tells a realistic growth story: near-breakeven in Q2 2025, growing to 25% margin by Q1 2026.

Founded	2019 · Series A
Offices	San Jose CA · Hyderabad IN
Revenue model	Platform license + services
Peer companies	Weights & Biases · Arize AI · Valohai

SEED SCRIPT · 207 EXPENSE ENTRIES

Salary · Software (AWS, Salesforce, Datadog) · Office (SJ + Hyderabad) · Marketing (re:Invent, NeurIPS, Gartner) · Travel · Legal

PRICING · INR ENTERPRISE MARKET

DKubeX annual ₹2.5 Cr · MLOps annual ₹1.5 Cr · Prof. services ₹25 L/mo · Blueprints ₹50–80 L

RESET · RE-SEED ANY TIME

```
python scripts/seed_data.py --reset
```

V1.2 DELIVERED • WHAT'S NEXT

All 11 features shipped. Roadmap ahead.

V1.2 – DELIVERED ✓

- ✓ AI expense auto-categorisation
- ✓ Receipt OCR (vision API)
- ✓ Invoice reminder drafts
- ✓ 12-month trend chart
- ✓ Statistical cash flow forecast
- ✓ Anomaly detection + LLM explanation
- ✓ AI monthly executive report
- ✓ Line-item suggestions
- ✓ NL financial Q&A (tool-use chat)
- ✓ Budget planning + AI guardrails
- ✓ Client profitability scoring

V2.0 – CANDIDATE BACKLOG

- **Streaming SSE chat**
Replace synchronous tool-use with server-sent events for real-time responses.
- **Multi-currency support**
USD/EUR/INR with real-time FX rates and currency-aware P&L.
- **Tally / QuickBooks export**
One-click export to Indian and global accounting software.
- **Investor-grade board pack**
Auto-generated monthly PDF board report with variance analysis.



DClaw Finance

Financial intelligence, built on One Convergence design.
Powered by DKube private AI.

₹39 Cr

Annual revenue (demo)

11

AI features shipped

24.7%

Net margin

12

Enterprise clients

- localhost:3007
- github.com · dclaw/dclaw-finance
- Docker · Kubernetes · Helm